

Chemistry reference

Melting and boiling points

Every element has a specific melting and boiling point. This is the temperature at which a solid changes into a liquid or a gas respectively. All temperatures are measured at atmospheric pressure. Metals tend to have high melting points, while simple gases have boiling points below room temperature. However, carbon is a nonmetal, but has the highest melting point of all.

LIST OF ELEMENTS					
Atomic number	Name/Symbol	Melting point °C °F		Boiling point °C °F	
1	hydrogen (H)	−259	−434	−253	−423
2	helium (He)	−272	−458	−269	−452
3	lithium (Li)	179	354	1,340	2,440
4	beryllium (Be)	1,283	2,341	2,990	5,400
5	boron (B)	2,300	4,170	3,660	6,620
6	carbon (C)	3,500	6,332	4,827	8,721
7	nitrogen (N)	−210	−346	−196	−321
8	oxygen (O)	−219	−362	−183	−297
9	fluorine (F)	−220	−364	−188	−306
10	neon (Ne)	−249	−416	−246	−410
11	sodium (Na)	98	208	890	1,634
12	magnesium (Mg)	650	1,202	1,105	2,021
13	aluminum (Al)	660	1,220	2,467	4,473
14	silicon (Si)	1,420	2,588	2,355	4,271
15	phosphorus (P)	44	111	280	536
16	sulfur (S)	113	235	445	832
17	chlorine (Cl)	−101	−150	−34	−29
18	argon (Ar)	−189	−308	−186	−303
19	potassium (K)	64	147	754	1,389
20	calcium (Ca)	848	1,558	1,487	2,709
21	scandium (Sc)	1,541	2,806	2,831	5,128
22	titanium (Ti)	1,677	3,051	3,277	5,931
23	vanadium (V)	1,917	3,483	3,377	6,111
24	chromium (Cr)	1,903	3,457	2,642	4,788
25	manganese (Mn)	1,244	2,271	2,041	3,706
26	iron (Fe)	1,539	2,802	2,750	4,980
27	cobalt (Co)	1,495	2,723	2,877	5,211
28	nickel (Ni)	1,455	2,641	2,730	4,950

LIST OF ELEMENTS					
Atomic number	Name/Symbol	Melting point		Boiling point	
		°C	°F	°C	°F
29	copper (Cu)	1,083	1,981	2,582	4,680
30	zinc (Zn)	420	788	907	1,665
31	gallium (Ga)	30	86	2,403	4,357
32	germanium (Ge)	937	1,719	2,355	4,271
33	arsenic (As)	817	1,503	613	1,135
34	selenium (Se)	217	423	685	1,265
35	bromine (Br)	−7	19	59	138
36	krypton (Kr)	−157	−251	−152	−242
37	rubidium (Rb)	39	102	688	1,270
38	strontium (Sr)	769	1,416	1,384	2,523
39	yttrium (Y)	1,522	2,772	3,338	6,040
40	zirconium (Zr)	1,852	3,366	4,377	7,911
41	niobium (Nb)	2,467	4,473	4,742	8,568
42	molybdenum (Mo)	2,610	4,730	5,560	10,040
43	technetium (Tc)	2,172	3,942	4,877	8,811
44	ruthenium (Ru)	2,310	4,190	3,900	7,052
45	rhodium (Rh)	1,966	3,571	3,727	6,741
46	palladium (Pd)	1,554	2,829	2,970	5,378
47	silver (Ag)	962	1,764	2,212	4,014
48	cadmium (Cd)	321	610	767	1,413
49	indium (In)	156	313	2,028	3,680
50	tin (Sn)	232	450	2,270	4,118
51	antimony (Sb)	631	1,168	1,635	2,975
52	tellurium (Te)	450	842	990	1,814
53	iodine (I)	114	237	184	363
54	xenon (Xe)	−112	−170	−107	−161
55	cesium (Cs)	29	84	671	1,240
56	barium (Ba)	725	1,337	1,640	2,984